Timing of Variable-Star Extrema Using Gaussian Process Regression

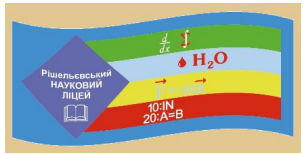
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Why precise timing matters

Lightcurves encode the behaviour of variable stars:

- stellar activity
- periodicity and period changes
- orbital dynamics
- transient events
- …

Precise timing of lightcurve extrema lets us detect notable effects.

High-cadence, long-term TESS observations are excellent material for this

Approaches to timing lightcurve extrema

Local modelling

- **Phenomenological**
  - simple analytical functions – parabola, Gaussian, …
  - shape-aware functions – e.g., MAVKA (Andrych & Andronov, 2019)
- **Semi-physical models** – e.g., (Tkachenko & Andronov, 2014)
- **Probabilistic / non-parametric** – Gaussian Process Regression

Template fitting

- **Empirical templates**
  - phase-averaged cycles of the same star – (Kato et al., 2013; this work)
  - high-quality reference lightcurve (Kato et al., 2009)
- **Physical templates**
  - PHOEBE-derived template (Zasche, P. et al., 2014)

Gaussian Process Regression (GP)

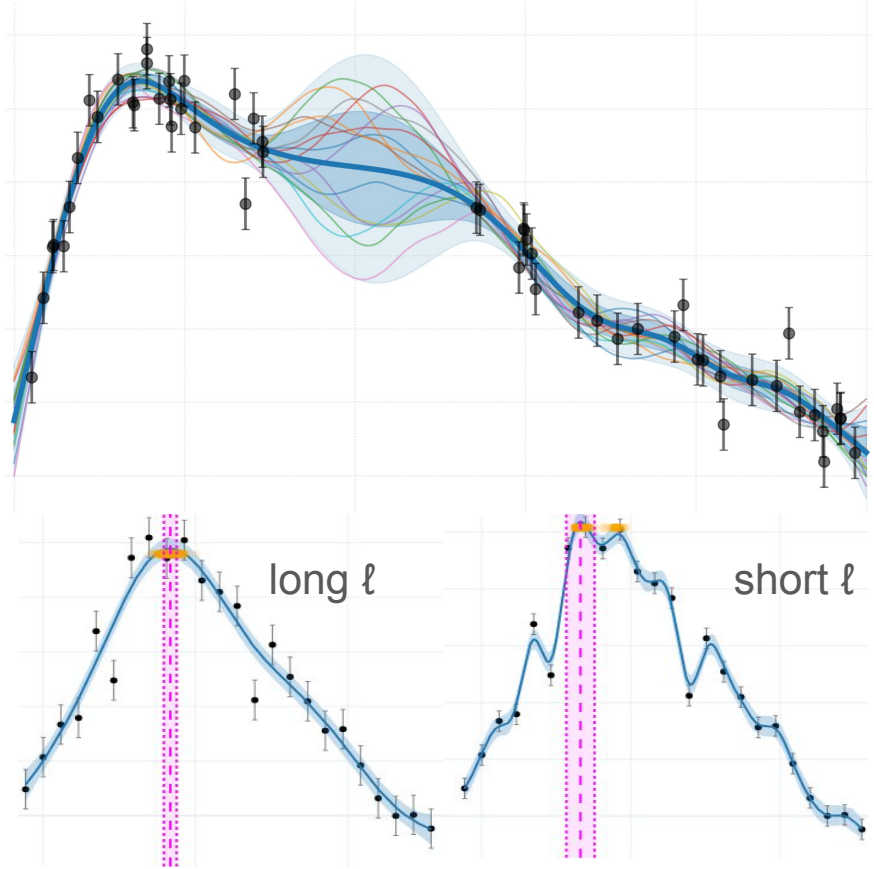
GP is a *probability distribution over possible curves*

GP is described by a *kernel* and its *hyperparameters*

Length-scale ℓ - how far correlations extend

Prior – family of possible curves defined by the selected kernel + hyperparameters

Posterior – prior, constrained by the observed data and their uncertainties



Gaussian Process: under the hood

$$k(x_i, x_j) = A^2 \exp\left(-\frac{(x_i - x_j)^2}{2\ell^2}\right) \quad \Rightarrow \quad K_{ij} = k(x_i, x_j) \quad \Rightarrow \quad \mathbf{f} \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$$

A kernel (e.g., RBF) $\Rightarrow$ covariance matrix $\Rightarrow$ Joint Gaussian distribution of function values

Python implementation: scikit-learn

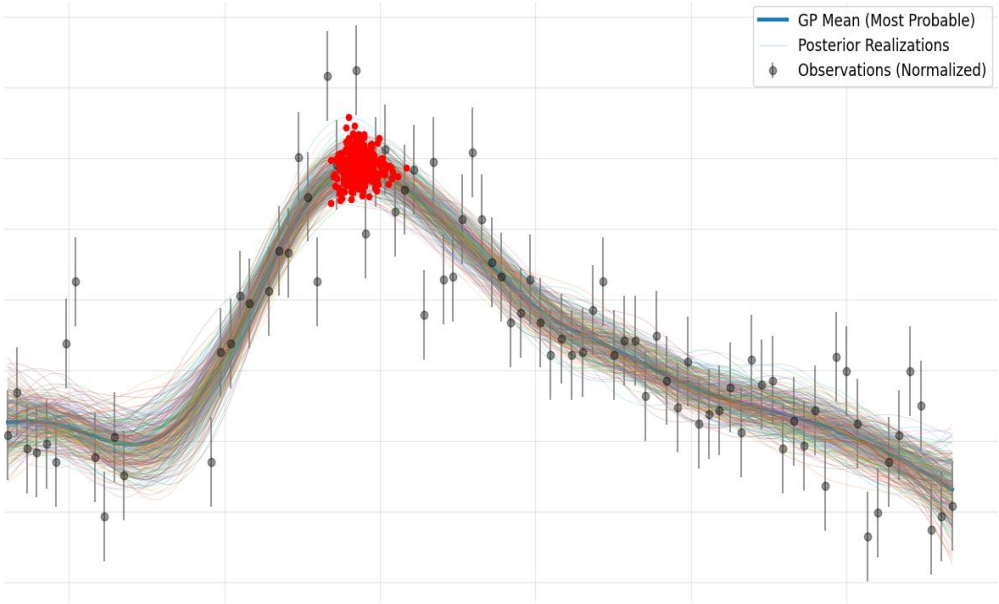


Gaussian Process for Timing Extrema

GP prediction -> **mean curve** + uncertainty

Mean curve -> **ToM** (extremum time)

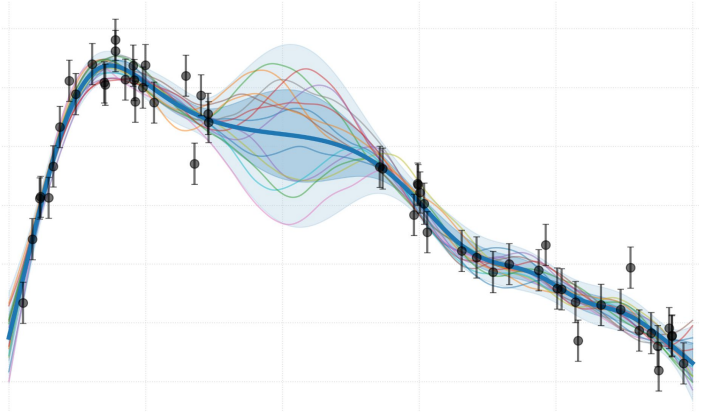
300 posterior realisations -> extremum times -> standard deviation -> **ToM uncertainty**



GP for lightcurves

GP is

- Shape-agnostic
- Handles:
  - irregularly sampled data
  - gaps (returns towards the prior mean, uncertainties increase)
- A probabilistic model of the light curve



Using GP to model variable star lightcurve is not new

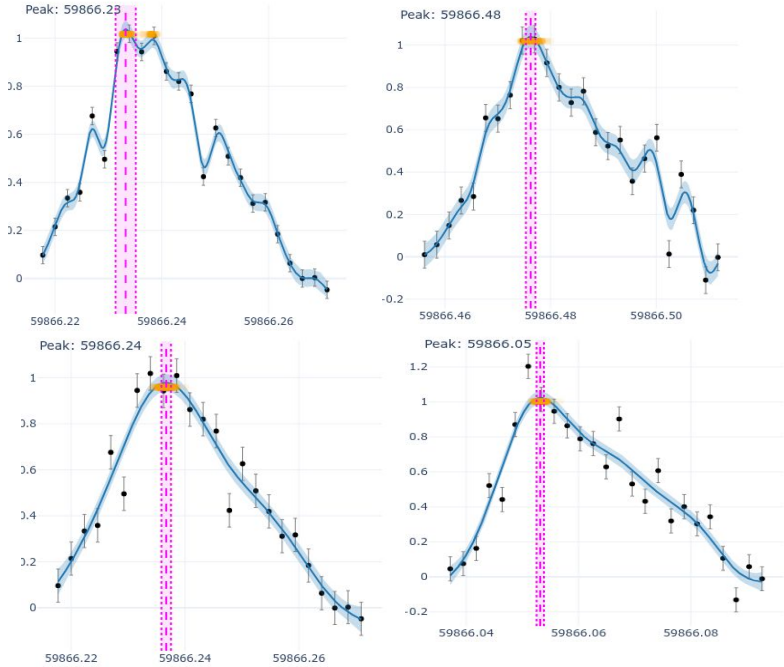
For example, lightcurve modelling for variable star classification:
(Castro et al., 2018; Trevisan et al., 2023)

Tuning the fit: Expert in the loop

Hyperparameters (*ideally*) are optimised from the data

In reality: in sparse data, it makes sense to borrow hyperparameters from well-covered cycles

Another, not quite fair but effective, tweak: rescale the noise

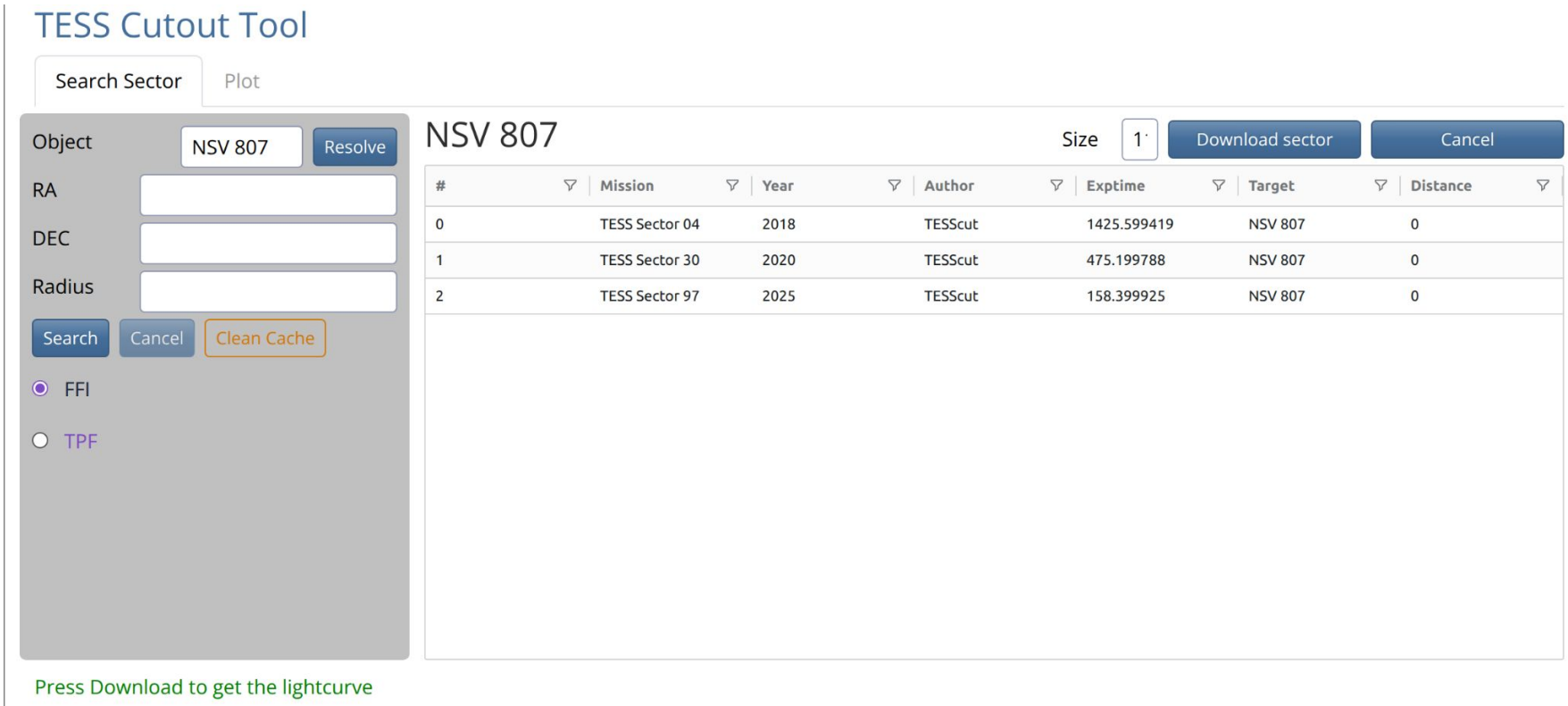


scale uncertainties

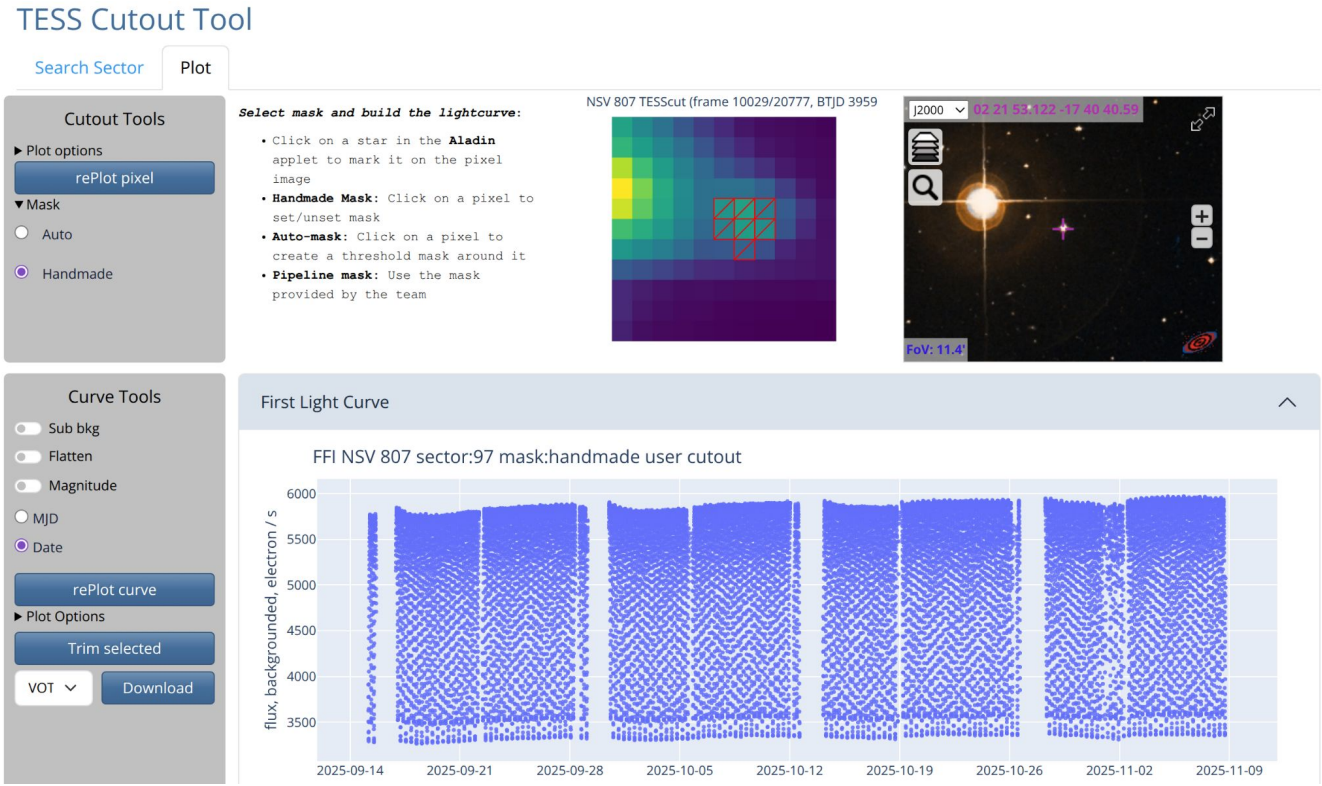
borrow length-scale

Workflow Demonstration

NSV 807 TESS Sector 97: Search for FFI

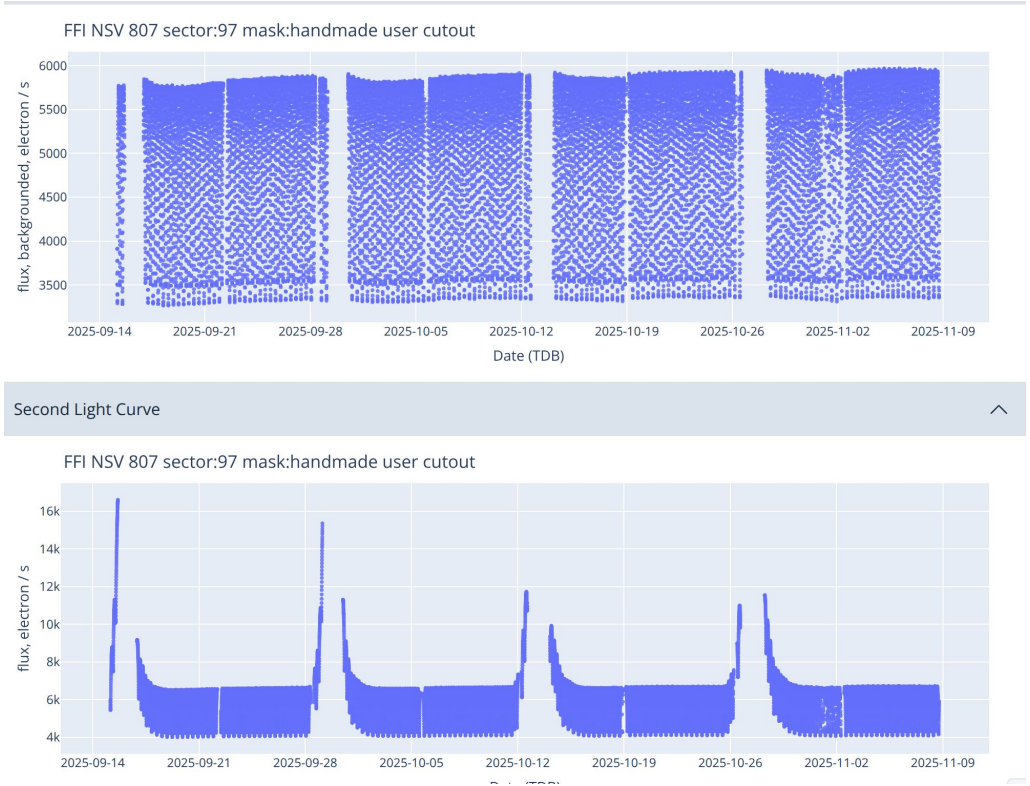


NSV 807 TESS Sector 97: Build the lightcurve

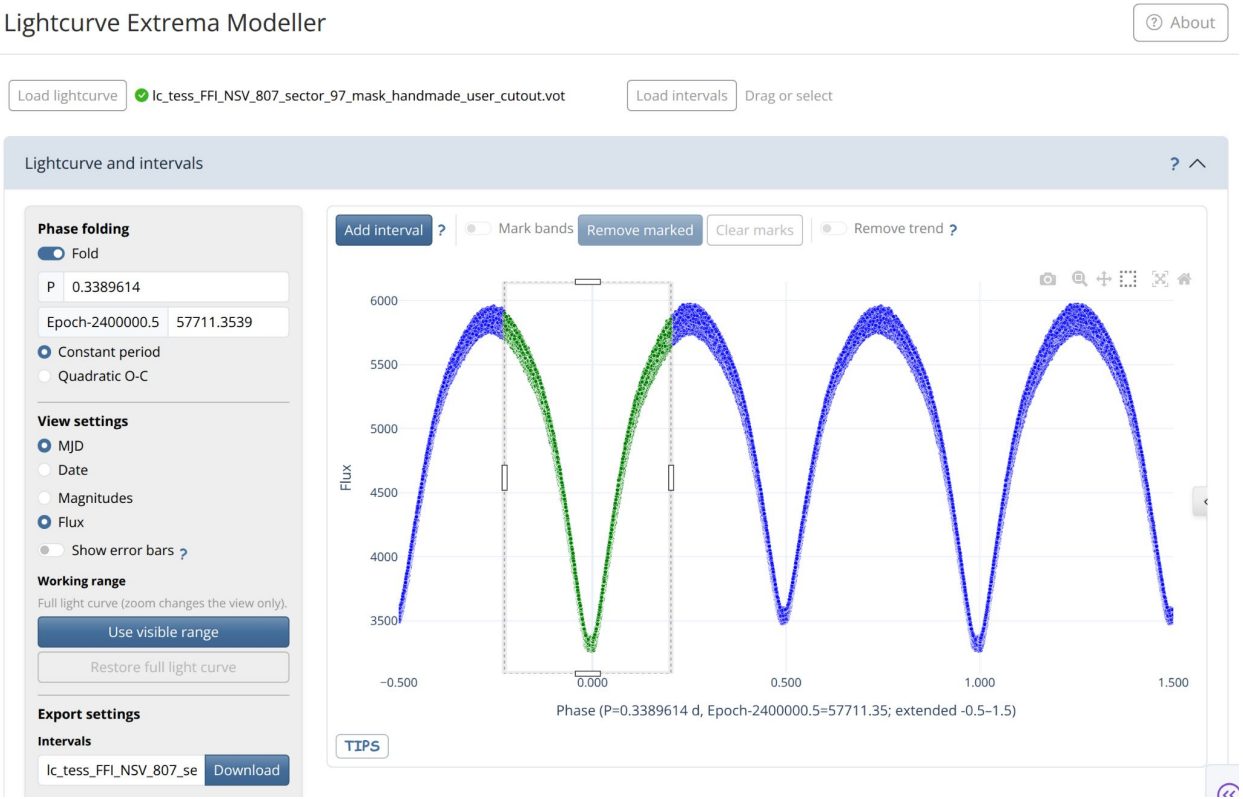


TESS lightcurve: Attention to the background

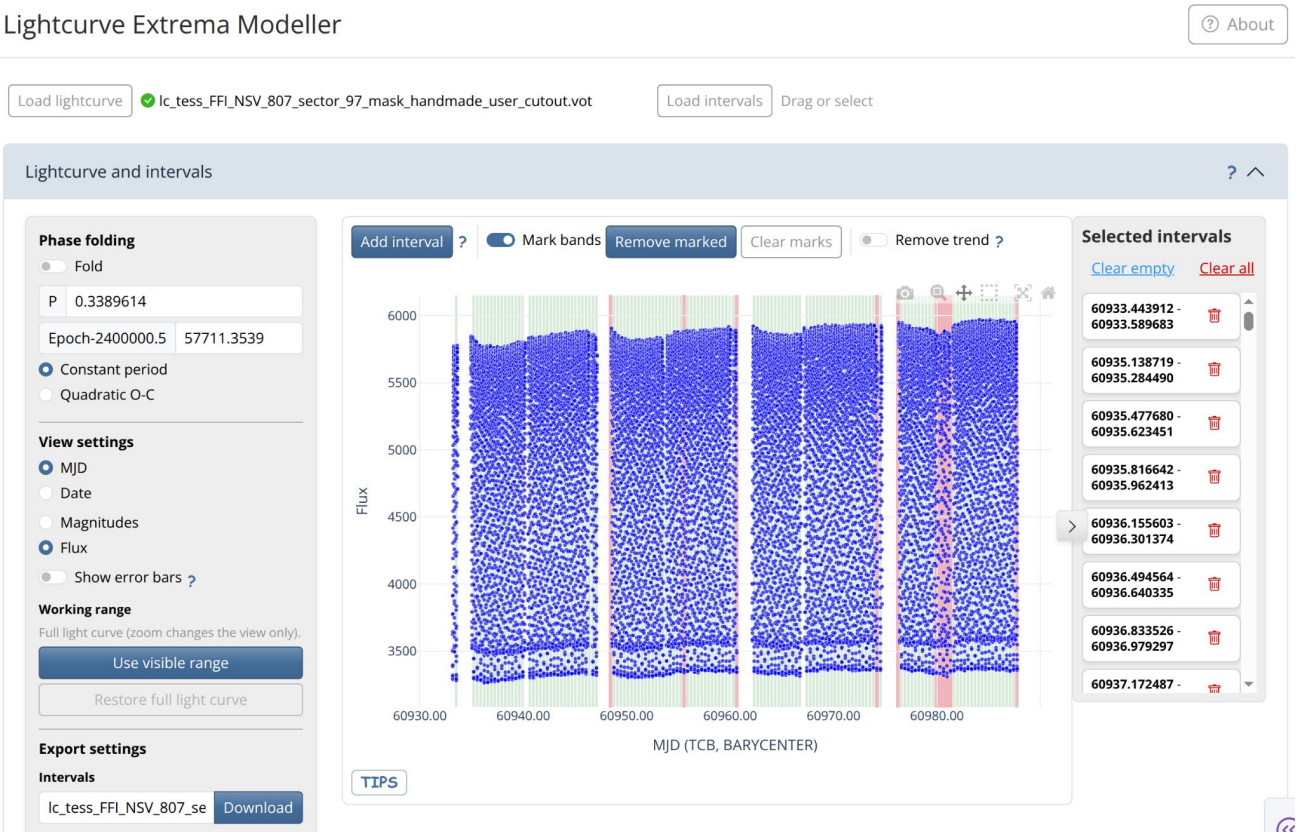
Background removal “improves” the lightcurve but may distort timing



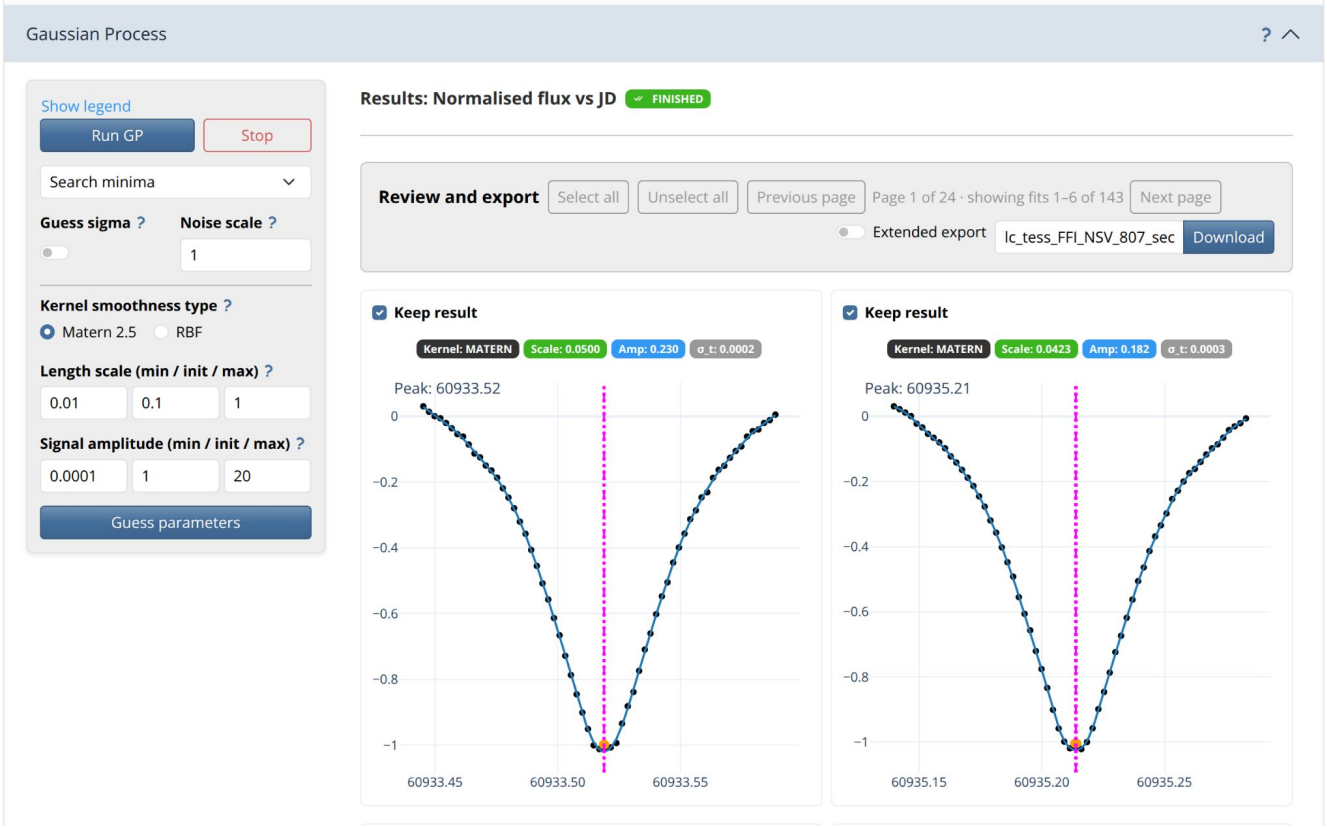
Extrema Modeller: Select intervals



Extrema Modeller: Select intervals



Extrema Modeller: Fit individual intervals



Observed – Calculated (O–C)



Observed – Calculated (O–C)



GP vs. MAVKA

Selected algorithms from **MAVKA** (Andrych & Andronov, 2019)

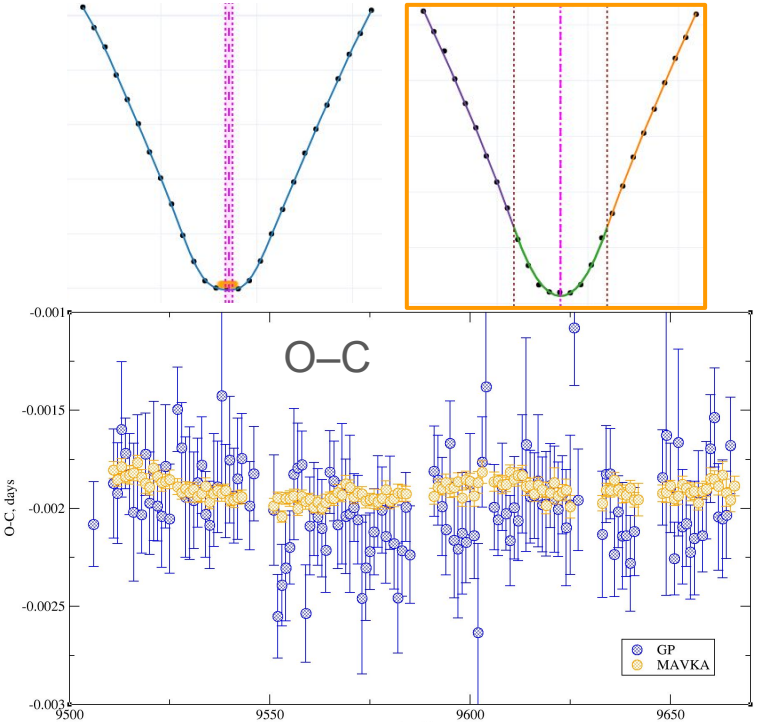
Algorithms developed for precise, shape-aware fitting of eclipsing binary lightcurves:

WSAP – Wall-Supported Asymptotic Parabola
near-flat-bottom minima, e.g. circular eclipses with limb darkening

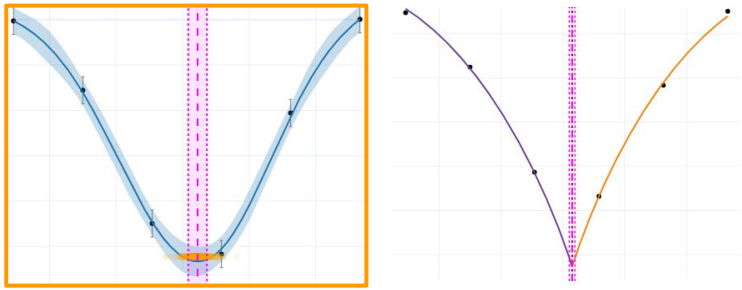
WSL – Wall-Supported Line
genuinely flat-bottom minima (total eclipses)

NSV 807: comparing two TESS cadences

Sector 97 (exposition 158s)

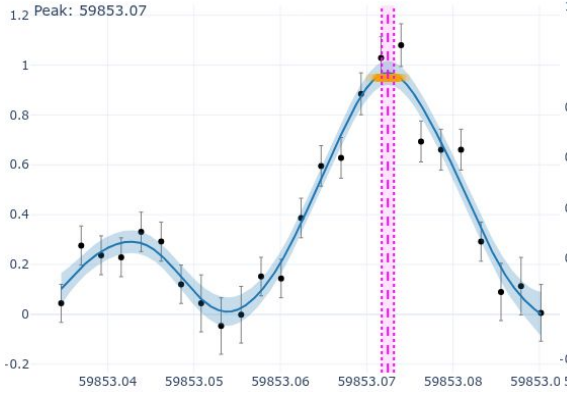


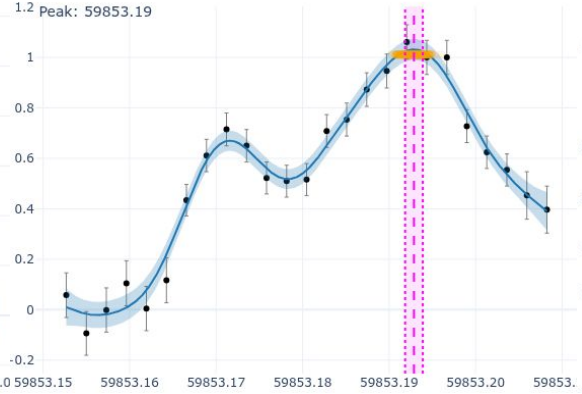
Sector 4 (exposition 1425s)

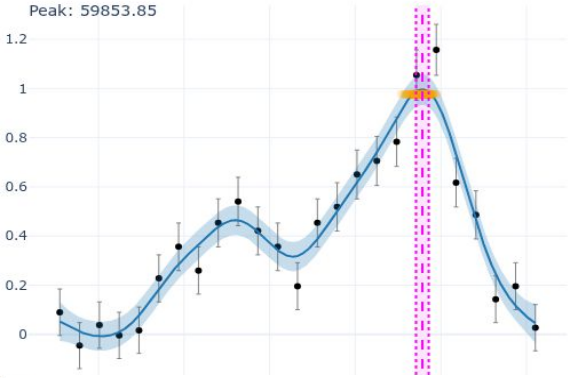


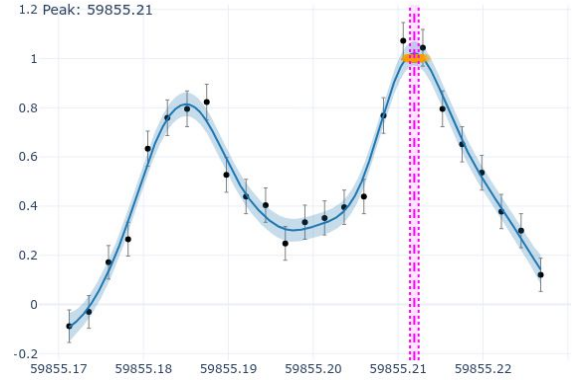
Robustness vs. precision
…but we don’t miss cycles

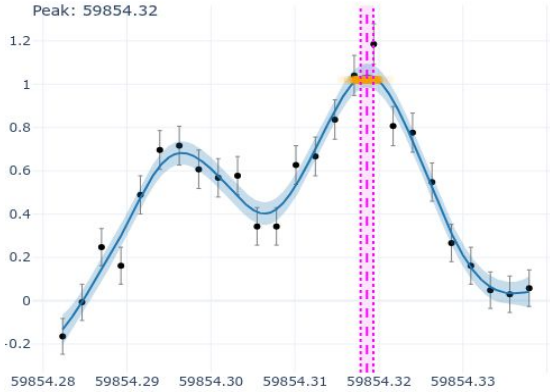
TCP J23580961+5502508: a superhump zoo

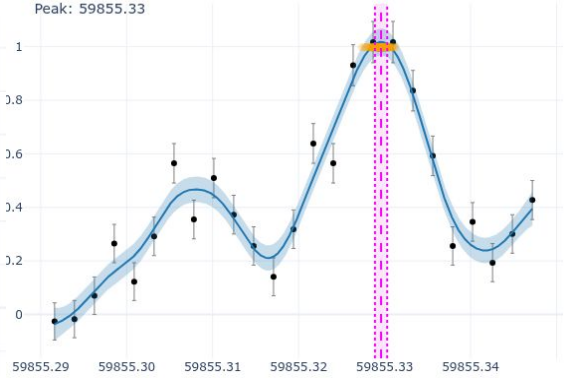












Who wins? It depends

lightcurve quality	GP	MAVKA	template shifting
dense, shaped	no shape prior→ less precise	shape-aware→ more precise	more human involvement
sparse, shaped	no shape prior	shape-aware, but may fail	precise on stable shapes
crazy, unstable	robust, shape-agnostic	implemented tools does not fit complex shape	risky (e.g. artificial phase shits)

Conclusions

Experimenting with algorithms is fun

Precision when possible; robustness when needed

More tools – more options - more freedom

…and an expert-in-the-loop

A Fair Question

Why are we bothering with individual stars in the *Big-Data era*?

There are more than 2 million eclipsing binaries in Gaia DR3…



- ***Curiosity***: individual stars can be peculiar or show unusual behaviour.
  And we need tools and algorithms to discover them in the vast datasets (manually, semi-automatically or automatically)

- Individual stars are our laboratory *for* ***scalable methods***
- **...**

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